# Propagated class exclusion for direct semi-supervised non-negative matrix factorization

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**Abstract**

Semi-supervised non-negative matrix factorization (NMF) commonly uses labels to shape a representation before a separate class decision. We formulate propagated class-exclusion NMF (PCE-NMF), in which each factor component denotes a class and graph-derived harmonic evidence penalizes unsupported class contributions. Labeled samples retain exact class support, and a scale-invariant contribution energy assigns unlabeled samples directly from the factors. We evaluate the method with 20 paired splits on ten data sets against eight established methods and two mechanism controls. PCE-NMF exceeds a matched hard-exclusion factorization on all ten data sets, isolating the value of propagated exclusion within the factorization. Its mean accuracy difference from harmonic prediction alone is $-0.0006$, with data-set-level gains and losses. PCE-NMF attains the highest accuracy among PCE-NMF, HLPNMF, and GOCNMF on COIL20, ORL, and COIL100, while HLPNMF leads on six data sets. A no-cache timing study places PCE-NMF between GOCNMF and HLPNMF on every data set. The study establishes a direct class-aware factorization and separates the contribution of propagated exclusion from the predictive strength of the underlying harmonic field. The source code is publicly available at https://github.com/matrix-research-lab-Wf/PCE-NMF.

*Keywords:* Non-negative matrix factorization, Semi-supervised clustering, Label propagation, Class exclusion, Direct clustering

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## 1. Introduction

Non-negative matrix factorization represents a non-negative data matrix through a small set of non-negative factors [1, 2]. In clustering, the sample factor indicates how strongly each sample is associated with each component. NMF has been used for document clustering [3]; its connections to spectral and orthogonal clustering formulations have also been studied [4, 5]. Sparsity constraints [6] and alternative numerical solvers [7, 8] address different aspects of the factorization. Limited class labels can make components easier to interpret: they may shape a representation [9], constrain labeled coordinates, or propagate class information to unlabeled samples [10].

The class meaning of an NMF component is usually assigned after the factorization. With a small labeled set, it can instead be specified during fitting. Each of the $c$ components then denotes one class, and the labeled rows of the sample factor have exact class support. For an unlabeled sample, support alone leaves the relative cost of candidate classes unspecified. A harmonic field on the sample graph provides class evidence at these unlabeled vertices.

PCE-NMF uses the complement of that evidence as a class-specific penalty on each factor contribution. The same contribution energy is used for the final class decision. This gives the graph evidence a specific role in the factorization while preserving ordinary NMF reconstruction. The construction also makes the decision invariant to positive diagonal rescaling of the two factors.

We examine three questions. How much does propagated exclusion change a hard-support factorization? How do factor-based decisions compare with predictions from the harmonic field itself? How does the method compare with established semi-supervised NMF methods under matched splits and native prediction rules? Ten-data-set experiments address these questions; additional fixed experiments measure sensitivity to label fraction, corruption, class count, runtime, and numerical convergence. The results show large gains over hard exclusion within the factorization, similar mean ACC to harmonic prediction, and strong performance on COIL20 and COIL100.

The paper makes three contributions. First, it defines a class-wise exclusion term on factor contribution energy and an aligned direct decision rule. Second, it establishes well-posedness of the regularized harmonic step, descent of the exact alternating updates under stated conditions, and scale invariance of class energy. Third, matched controls isolate the contribution of propagated exclusion and show that the resulting factor-based decisions have data-set-

specific gains and losses relative to direct harmonic prediction. A nine-method benchmark and no-cache timing study characterize the predictive and computational behavior of the construction.

## 2. Related Work

### *2.1. Representation-level semi-supervision in NMF*

Constrained NMF incorporates partial labels into the factorization to learn a discriminative representation [9]. Other representation-level methods use discriminative terms on labeled samples [11, 12], local constraints [13], or a combination of label discrimination and graph regularization [14]. The learned coordinates can then support a downstream class decision. PCE-NMF uses $c$ components with explicit class meanings and restricts the labeled rows of $V$ to their known class support. Its unlabeled penalty acts on the contribution of each class component to each sample, so the class decision is obtained from the factors themselves.

### *2.2. Graph and propagation mechanisms*

Graph-based clustering includes normalized cuts [15], Laplacian embeddings [16], and spectral clustering [17]. Graph-regularized NMF preserves local relations in the low-dimensional representation [18]; symmetric NMF provides another factorization of a sample affinity matrix [19]. Harmonic label propagation solves a graph boundary-value problem to extend labels from observed to unobserved vertices [10], while local and global consistency uses a graph-smoothing formulation [20].

In semi-supervised NMF, pairwise constraints can modify the graph [21]. Label and constraint propagation have also been combined in the factorization [22]; hypergraph methods use higher-order sample relations [23], and pivotal-aware models propagate both point labels and pairwise constraints [24]. Weighted label propagation is coupled to NMF in SNMFWLP [25]. HLPNMF couples high-order propagation with NMF through a hypergraph-induced transition structure [26]. PCE-NMF computes a harmonic class field first and uses it as fixed class-exclusion evidence during factorization. Thus its graph affects the class penalties, while the reconstruction remains the ordinary two-factor NMF term.

### 2.3. Direct class-semantic factorization

GOCNMF embeds one-hot labels in the representation of labeled samples and applies graph regularization to guide unlabeled representations [27]. PCE-NMF also assigns class meanings to factor coordinates, but it uses hard class support rather than fixing labeled coordinates to one-hot values. For unlabeled samples, it penalizes class-specific reconstruction energy according to harmonic evidence. This separates the observed label constraint from the propagated preference over candidate classes. Robust semi-supervised NMF [28] and self-supervised semi-supervised NMF [29] pursue complementary goals through robust losses or additional self-supervisory signals.

Table 1 locates PCE-NMF relative to the closest compared methods. Its defining choice is to apply propagated evidence to the scale-invariant energy of each class contribution, followed by a decision on that same energy.

Table 1: How closely related methods use labels and make class decisions. The entries describe the formulations and native prediction rules used in the paired comparison.

| Method | Role of supervision in the factorization | Native decision |
|---|---|---|
| GOCNMF | One-hot labeled coordinates and graph regularization of the representation | Maximum representation coordinate |
| SNMFWLP | Weighted label propagation coupled to NMF with basis orthogonality | K-means on the representation |
| HLPNMF | High-order propagation regularization through a shared coefficient matrix | Maximum class-semantic coordinate |
| PCE-NMF | Harmonic evidence weights class-specific contribution energy under exact labeled support | Maximum contribution energy |

## 3. Propagated Class-Exclusion NMF

### 3.1. Problem formulation

Let $X = [x_1, \ldots, x_n] \in \mathbb{R}_+^{d\times n}$ contain $n$ non-negative samples with $d$ features, and let $c$ be the number of classes. We factorize $X \approx UV^\top$ with $U = [u_1, \ldots, u_c] \in \mathbb{R}_+^{d\times c}$ and $V = (v_{ik}) \in \mathbb{R}_+^{n\times c}$. A labeled index set $\mathcal{L} \subseteq \{1, \ldots, n\}$ supplies labels $y_i \in \{1, \ldots, c\}$ for $i \in \mathcal{L}$. The class identity of component $k$ is fixed by these labels. Nonzero samples are normalized to unit Euclidean norm before graph construction and factorization.

*3.2. Harmonic class evidence*

Construct a binary $p$-nearest-neighbor graph from Euclidean distances between the normalized samples and symmetrize its adjacency matrix $W \in \{0,1\}^{n\times n}$. Let $D = \mathrm{diag}(W\mathbf{1})$ and $L = D - W$. For each connected component containing both labeled and unlabeled vertices, partition its Laplacian by the labeled and unlabeled indices. The labeled one-hot matrix $C$ defines the regularized harmonic field

$$(L_{\mathcal{U}\mathcal{U}} + \varepsilon I)F_{\mathcal{U}} = W_{\mathcal{U}\mathcal{L}}C, \qquad \varepsilon = 10^{-12}. \tag{1}$$

The implementation clips negative numerical entries to zero and normalizes each nonzero row to sum to one. A component without labeled vertices contributes no propagated exclusion. Let $P \in [0,1]^{n\times c}$ denote the resulting class evidence on anchored unlabeled vertices; the known one-hot labels are retained on $\mathcal{L}$.

*3.3. Propagated exclusion model*

Define the exclusion weights $Q \in [0,1]^{n\times c}$ by

$$Q_{ik} = \begin{cases} 0, & i \in \mathcal{L},\ k = y_i, \\ 1, & i \in \mathcal{L},\ k \neq y_i, \\ 1 - P_{ik}, & i \notin \mathcal{L} \text{ in an anchored component}, \\ 0, & i \notin \mathcal{L} \text{ in an unanchored component}. \end{cases} \tag{2}$$

The PCE-NMF objective is

$$\min_{U,V\geq 0} \quad \mathcal{J}(U,V) = \|X - UV^\top\|_F^2 + \lambda \sum_{i=1}^{n}\sum_{k=1}^{c} Q_{ik}\|u_k\|_2^2 v_{ik}^2, \tag{3}$$

Here $\lambda \geq 0$ controls exclusion strength. The weight $1 - P_{ik}$ penalizes a class contribution more strongly when the graph provides less evidence for that class. The hard labeled-support constraint $v_{ik} = 0$ for $i \in \mathcal{L}$ and $k \neq y_i$ remains exact throughout optimization.

*3.4. Direct clustering rule*

After optimization, sample $i$ receives the class

$$\hat{y}_i = \arg\max_{1\leq k\leq c} s_{ik}, \qquad s_{ik} = \|u_k\|_2^2 v_{ik}^2. \tag{4}$$

This rule uses the class meaning of the factors directly and assigns samples from $V$ in one step. Figure 1 summarizes the path from labeled samples through graph-derived exclusion to the class decision.

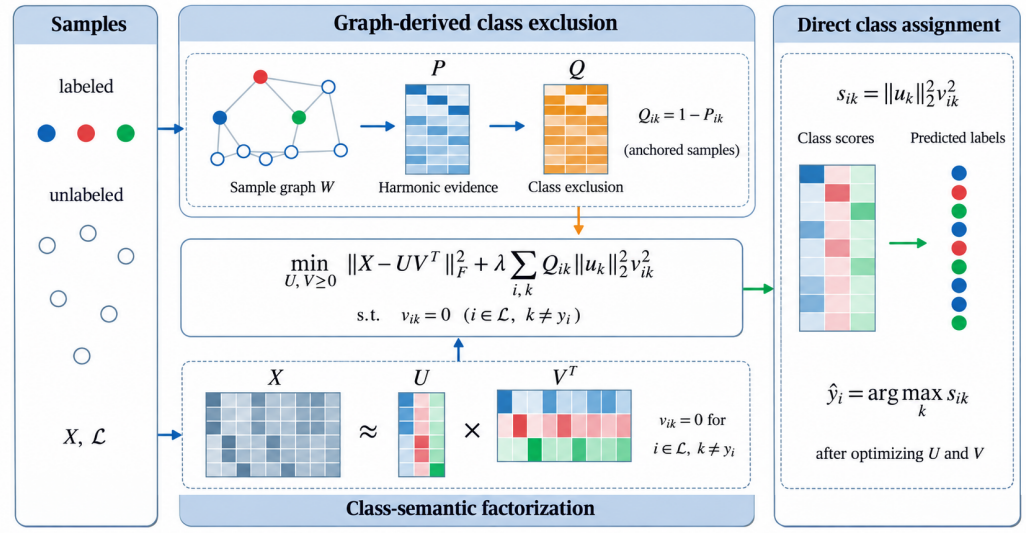



Figure 1: PCE-NMF mechanism. Labeled samples anchor harmonic class evidence $P$ on the sample graph; the resulting exclusion weights $Q$ guide class-semantic factorization. After optimization, contribution energies give direct class assignments. The heatmaps and sample colors are schematic.

### *3.5. Optimization*

Write $a_k = \|u_k\|_2^2$ and $b_k = \sum_i Q_{ik} v_{ik}^2$. The gradients of the objective are

$$\nabla_U \mathcal{J} = 2\Big(UV^\top V - XV + \lambda U \operatorname{diag}(b)\Big), \tag{5}$$

$$\nabla_V \mathcal{J} = 2\Big(VU^\top U - X^\top U + \lambda (Q \odot V) \operatorname{diag}(a)\Big). \tag{6}$$

The implemented alternating multiplicative updates are

$$U \leftarrow U \odot \frac{XV}{UV^\top V + \lambda U \operatorname{diag}(b)}, \tag{7}$$

$$V \leftarrow V \odot \frac{X^\top U}{VU^\top U + \lambda (Q \odot V) \operatorname{diag}(a)}. \tag{8}$$

The fractions are elementwise. Denominators are bounded below by $10^{-12}$ in the MATLAB implementation. After each update, forbidden labeled coordinates are reset to zero. The formal protocol uses 50 alternating updates and the parameters locked before evaluation.

*Algorithm 1: PCE-NMF optimization..*

1. **Input:** Non-negative data $X$, labeled indices $\mathcal{L}$ and labels $y_i$, class count $c$, graph size $p$, locked penalty $\lambda$, and seed. Set $T = 50$.
2. Normalize nonzero columns of $X$; construct the symmetric binary $p$-nearest-neighbor graph $W$. In each anchored component solve Eq. (1), normalize its nonzero rows, and form $Q$ by Eq. (2).
3. Initialize non-negative $U$ and $V$ from the recorded seed. Set $v_{ik} = 0$ for labeled $i$ and $k \neq y_i$.
4. For $t = 1, \ldots, T$, update $U$ and then $V$ by Eqs. (7)–(8); bound denominators below by $10^{-12}$ and reset forbidden labeled entries of $V$ to zero.
5. **Output:** Factors $U, V$ and unlabeled assignments from Eq. (4).

### *3.6. Computational complexity*

The blocked distance calculation for the graph requires $O(dn^2)$ arithmetic; selecting neighbors by sorting each distance column adds $O(n^2 \log n)$ comparisons. The symmetric binary graph contains $O(np)$ edges. The harmonic step solves one sparse system per anchored connected component; its time and memory depend on sparse factorization fill-in. Each factor update costs $O(dnc + (d+n)c^2)$ arithmetic and stores $O((d+n)c)$ factor entries, in addition to $X$ and the graph. These bounds describe the implementation's main operations; measured wall times are reported in Section 5.

## 4. Theoretical Analysis

The descent proof uses the auxiliary-function approach for multiplicative NMF updates [2, 26] and accounts for the class-exclusion terms in both factor blocks.

**Proposition 1** (Well-posed regularized harmonic step)**.** *For an undirected graph with non-negative weights and $\varepsilon > 0$, the matrix $L_{\mathcal{U}\mathcal{U}} + \varepsilon I$ in Eq.* (1) *is symmetric positive definite. Hence the regularized harmonic field is uniquely defined for every right-hand side.*

*Proof.* The graph Laplacian is symmetric positive semidefinite because

$$z^\top L z = \tfrac{1}{2} \sum_{i,j} W_{ij}(z_i - z_j)^2 \geq 0$$

for every $z \in \mathbb{R}^n$. A principal submatrix of a positive semidefinite matrix is positive semidefinite. Therefore, for nonzero $h \in \mathbb{R}^{|\mathcal{U}|}$, $h^\top (L_{\mathcal{U}\mathcal{U}} + \varepsilon I)h \geq \varepsilon \|h\|_2^2 > 0$. Symmetric positive definiteness gives invertibility and a unique solution. □

**Theorem 1** (Descent of the exact alternating updates)**.** *Fix $Q \in [0,1]^{n \times c}$, let $X \in \mathbb{R}_+^{d \times n}$, and take $\lambda \geq 0$. Apply Eqs.* (7)–(8) *in exact arithmetic, keeping forbidden labeled coordinates at zero. At each update, assume every positive, permitted coordinate has a positive denominator; any coordinate that reaches zero remains zero. Each factor update then does not increase $\mathcal{J}$. Consequently, the exact alternating sequence has nonincreasing objective values that converge to a finite limit.*

*Proof.* For fixed $V$, let $u \in \mathbb{R}_+^c$ be one row of $U$ and $x$ the corresponding row of $X$. Up to a constant, its objective is $f(u) = u^\top A u - 2h^\top u$, where $A = V^\top V + \lambda \operatorname{diag}(b)$ and $h = xV$. For fixed $U$, the corresponding expression for row $v_i$ of $V$ has $A_i = U^\top U + \lambda \operatorname{diag}(Q_{i:} \odot a)$ and $h_i = x_i^\top U$, where $x_i$ is column $i$ of $X$. Both versions of $A$ are symmetric and entrywise nonnegative, and their corresponding $h$ are nonnegative. Restrict a row to its positive, permitted coordinates, write the current vector as $z > 0$, and set $D = \operatorname{diag}((Az)_k / z_k)$. For any $\delta$,

$$\delta^\top (D - A)\delta = \tfrac{1}{2} \sum_{j,k} A_{jk} z_j z_k \left( \frac{\delta_j}{z_j} - \frac{\delta_k}{z_k} \right)^2 \geq 0.$$

Hence $g(z + \delta \mid z) = f(z) + 2(Az - h)^\top \delta + \delta^\top D \delta$ majorizes $f(z + \delta)$ and is equal to it at $\delta = 0$. The positive diagonal matrix $D$ gives the nonnegative minimizer $z^+ = D^{-1}h = z \odot h/(Az)$, which is the multiplicative update on these coordinates. Thus $f(z^+) \leq g(z^+ \mid z) \leq g(z \mid z) = f(z)$. If a numerator vanishes, the updated coordinate becomes zero and is held there; the same argument applies to the remaining active coordinates. Summing over rows proves descent for each factor block. Finally, $\mathcal{J} \geq 0$, so its sequence of objective values converges to a finite limit. □

Theorem 1 establishes convergence of objective values for the exact updates. The MATLAB implementation bounds denominators below by $10^{-12}$ and clips factor entries to that level; its objective trajectories are measured in Section 5.

**Proposition 2** (Scale invariance of class energy)**.** *For any positive diagonal matrix* $T = \mathrm{diag}(t_1, \ldots, t_c)$*, the feasible transformation* $U \mapsto UT$ *and* $V \mapsto VT^{-1}$ *leaves the reconstruction, the exclusion penalty, and every class score in Eq.* (4) *unchanged.*

*Proof.* Since $T$ is diagonal and positive, $(UT)(VT^{-1})^\top = UV^\top$. For each class $k$, $\|t_k u_k\|_2^2 (v_{ik}/t_k)^2 = \|u_k\|_2^2 v_{ik}^2$. This identity applies to each term of the exclusion penalty and to each class score. Multiplication by a positive diagonal matrix also preserves non-negativity and every forbidden zero in $V$. □

For an anchored unlabeled sample, $Q_{ik} = 1 - P_{ik}$ gives the exact identity

$$\sum_k Q_{ik} s_{ik} = \sum_k s_{ik} - \sum_k P_{ik} s_{ik}. \tag{9}$$

The penalty therefore reduces the cost of contributions aligned with the propagated evidence, relative to contributions to unsupported classes. Equation (9) is an algebraic interpretation of the fixed objective; the measured effect of this mechanism is tested by the matched controls in Section 5.

## 5. Experiments

### *5.1. Experimental settings*

We evaluated NMF, CNMF, GNMFLD, EDNMF, LCNMF, GOCNMF, SNMFWLP, HLPNMF, and PCE-NMF on YaleB, COIL20, Optdigits, ORL, PIE, COIL100, AR, MSRA25, MNIST, and USPS. COIL20 and COIL100 originate from the Columbia Object Image Library [30, 31]; MNIST is the handwritten-digit benchmark described by LeCun et al. [32]. These ten data sets were fixed before the formal comparison and cover face, object, and digit recognition tasks. Nonzero samples were normalized to unit Euclidean norm. For each class, $\max\{1, \lfloor 0.1 n_k \rfloor\}$ samples were labeled, where $n_k$ is the class size. All methods received the same labeled split in each of 20 paired runs. Accuracy (ACC), normalized mutual information (NMI), purity, F-score, precision, and recall were measured on unlabeled samples. The external methods retained their native training and prediction rules. Parameters were fixed before the formal runs using source settings where available and separate development splits for PCE-NMF.

Table 2 gives the actual matrix dimensions and labeled sample counts used in the formal comparison. The number of labeled samples follows the per-class floor rule above; all reported scores use the remaining samples.

Table 2: Data sets in the locked formal comparison. $n$ is the sample count, $d$ is the feature dimension, $c$ is the class count, and $n_{\mathcal{L}}$ is the number of labeled samples per paired split.

| Data set | $n$ | $d$ | $c$ | $n_{\mathcal{L}}$ |
|---|---|---|---|---|
| YaleB | 2414 | 1024 | 38 | 227 |
| COIL20 | 1440 | 1024 | 20 | 140 |
| Optdigits | 5620 | 64 | 10 | 557 |
| ORL | 400 | 4096 | 40 | 40 |
| PIE | 2856 | 1024 | 68 | 272 |
| COIL100 | 7200 | 1024 | 100 | 700 |
| AR | 1680 | 2000 | 120 | 120 |
| MSRA25 | 1799 | 256 | 12 | 173 |
| MNIST | 6996 | 784 | 10 | 695 |
| USPS | 9298 | 256 | 10 | 925 |

### *5.2. Clustering performance*

Table 3 summarizes the 20-seed means for PCE-NMF and two graph-based competitors. Tables 4– 13 immediately below report the complete nine-method, six-metric means and standard deviations for each data set. PCE-NMF achieved the highest ACC among these three methods on COIL20, ORL, and COIL100. HLPNMF led on YaleB, Optdigits, PIE, AR, MSRA25, and USPS, while GOCNMF led on MNIST. The pattern changes with the data set and metric: for example, PCE-NMF attained ACC/NMI of 0.9100/0.9452 on COIL100, whereas HLPNMF attained 0.8740/0.9229; on PIE the corresponding values were 0.9376/0.9527 and 0.9589/0.9714.

Figure 2 displays the mean ACC and NMI differences between PCE-NMF and each baseline across the ten data sets.

Table 3: Mean ACC and NMI over 20 paired seeds on unlabeled samples. Full nine-method results follow in this section.

| | ACC | | | NMI | | |
|---|---|---|---|---|---|---|
| Data set | PCE-NMF | HLPNMF | GOCNMF | PCE-NMF | HLPNMF | GOCNMF |
| YaleB | 0.6934 | 0.7308 | 0.7088 | 0.6979 | 0.7323 | 0.7054 |
| COIL20 | 0.9713 | 0.9515 | 0.9100 | 0.9725 | 0.9518 | 0.9032 |
| Optdigits | 0.9832 | 0.9859 | 0.9840 | 0.9579 | 0.9632 | 0.9593 |
| ORL | 0.6506 | 0.5935 | 0.6426 | 0.7910 | 0.7904 | 0.7963 |
| PIE | 0.9376 | 0.9589 | 0.9352 | 0.9527 | 0.9714 | 0.9534 |
| COIL100 | 0.9100 | 0.8740 | 0.8166 | 0.9452 | 0.9229 | 0.8731 |
| AR | 0.6332 | 0.6390 | 0.6323 | 0.8381 | 0.8418 | 0.8394 |
| MSRA25 | 0.9788 | 0.9941 | 0.9748 | 0.9607 | 0.9878 | 0.9588 |
| MNIST | 0.9315 | 0.9291 | 0.9352 | 0.8501 | 0.8508 | 0.8571 |
| USPS | 0.9700 | 0.9719 | 0.9695 | 0.9226 | 0.9268 | 0.9222 |

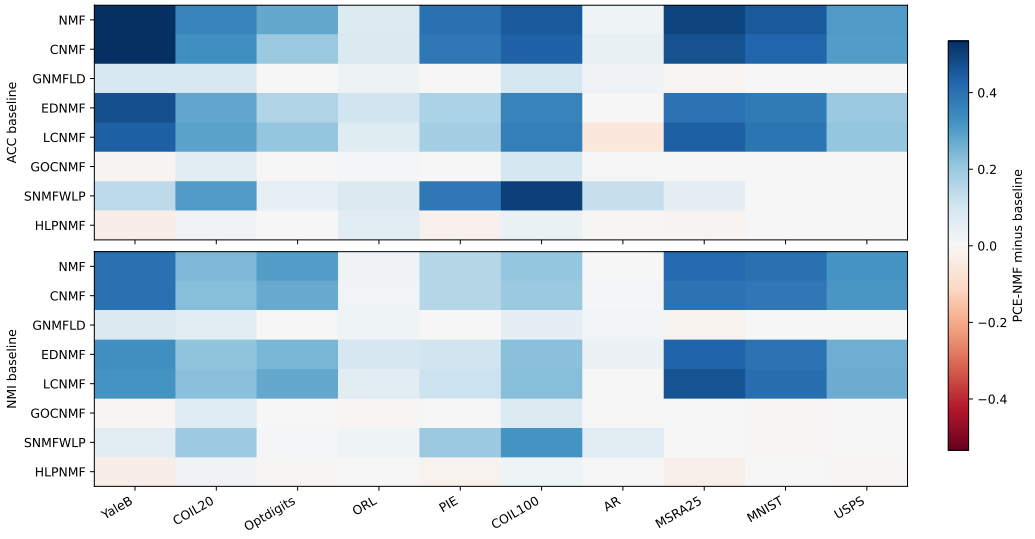



Figure 2: Differences in mean ACC and NMI between PCE-NMF and each baseline on the ten data sets. Each cell averages 20 matched seeds; positive values favor PCE-NMF.

Table 4: Complete formal results on YaleB. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

| Method | ACC | NMI | Purity | F-score | Precision | Recall |
|---|---|---|---|---|---|---|
| NMF | 0.1611 ± 0.0095 | 0.2993 ± 0.0143 | 0.1680 ± 0.0094 | 0.1023 ± 0.0066 | 0.0927 ± 0.0068 | 0.1142 ± 0.0069 |
| CNMF | 0.1594 ± 0.0080 | 0.2984 ± 0.0102 | 0.1670 ± 0.0087 | 0.1017 ± 0.0060 | 0.0915 ± 0.0062 | 0.1147 ± 0.0066 |
| GNMFLD | 0.6091 ± 0.0144 | 0.6203 ± 0.0105 | 0.6096 ± 0.0141 | 0.4087 ± 0.0159 | 0.3938 ± 0.0180 | 0.4250 ± 0.0143 |
| EDNMF | 0.2227 ± 0.0156 | 0.3717 ± 0.0125 | 0.2329 ± 0.0159 | 0.1411 ± 0.0142 | 0.1236 ± 0.0170 | 0.1674 ± 0.0117 |
| LCNMF | 0.2567 ± 0.0105 | 0.3807 ± 0.0075 | 0.2655 ± 0.0109 | 0.1463 ± 0.0057 | 0.1373 ± 0.0057 | 0.1565 ± 0.0060 |
| GOCNMF | 0.7088 ± 0.0131 | 0.7054 ± 0.0087 | 0.7089 ± 0.0129 | 0.5309 ± 0.0138 | 0.5173 ± 0.0142 | 0.5452 ± 0.0141 |
| SNMFWLP | 0.5544 ± 0.0210 | 0.6368 ± 0.0185 | 0.5689 ± 0.0202 | 0.2177 ± 0.0292 | 0.1439 ± 0.0271 | 0.4624 ± 0.0301 |
| HLPNMF | 0.7308 ± 0.0129 | 0.7323 ± 0.0095 | 0.7308 ± 0.0128 | 0.5651 ± 0.0155 | 0.5506 ± 0.0163 | 0.5804 ± 0.0151 |
| PCE_NMF | 0.6934 ± 0.0187 | 0.6979 ± 0.0142 | 0.6934 ± 0.0187 | 0.4952 ± 0.0255 | 0.4672 ± 0.0295 | 0.5271 ± 0.0212 |

Table 5: Complete formal results on COIL20. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

| Method | ACC | NMI | Purity | F-score | Precision | Recall |
|---|---|---|---|---|---|---|
| NMF | 0.6182 ± 0.0366 | 0.7323 ± 0.0159 | 0.6402 ± 0.0312 | 0.5606 ± 0.0281 | 0.5235 ± 0.0360 | 0.6042 ± 0.0234 |
| CNMF | 0.6419 ± 0.0327 | 0.7457 ± 0.0168 | 0.6614 ± 0.0272 | 0.5866 ± 0.0269 | 0.5562 ± 0.0313 | 0.6211 ± 0.0264 |
| GNMFLD | 0.8873 ± 0.0075 | 0.9146 ± 0.0076 | 0.8880 ± 0.0079 | 0.8478 ± 0.0105 | 0.8356 ± 0.0137 | 0.8605 ± 0.0115 |
| EDNMF | 0.6947 ± 0.0096 | 0.7594 ± 0.0096 | 0.6979 ± 0.0100 | 0.6178 ± 0.0115 | 0.6013 ± 0.0118 | 0.6354 ± 0.0119 |
| LCNMF | 0.6859 ± 0.0180 | 0.7492 ± 0.0137 | 0.6897 ± 0.0154 | 0.6059 ± 0.0186 | 0.5887 ± 0.0210 | 0.6242 ± 0.0163 |
| GOCNMF | 0.9100 ± 0.0198 | 0.9032 ± 0.0225 | 0.9100 ± 0.0198 | 0.8389 ± 0.0332 | 0.8281 ± 0.0355 | 0.8501 ± 0.0314 |
| SNMFWLP | 0.6682 ± 0.0379 | 0.7803 ± 0.0379 | 0.7031 ± 0.0361 | 0.5352 ± 0.0525 | 0.4531 ± 0.0546 | 0.6563 ± 0.0543 |
| HLPNMF | 0.9515 ± 0.0099 | 0.9518 ± 0.0093 | 0.9515 ± 0.0099 | 0.9110 ± 0.0174 | 0.9027 ± 0.0197 | 0.9196 ± 0.0155 |
| PCE_NMF | 0.9713 ± 0.0050 | 0.9725 ± 0.0040 | 0.9713 ± 0.0050 | 0.9473 ± 0.0089 | 0.9411 ± 0.0120 | 0.9536 ± 0.0066 |

Table 6: Complete formal results on Optdigits. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

| Method | ACC | NMI | Purity | F-score | Precision | Recall |
|---|---|---|---|---|---|---|
| NMF | 0.7106 ± 0.0372 | 0.6578 ± 0.0213 | 0.7281 ± 0.0314 | 0.6090 ± 0.0336 | 0.5894 ± 0.0399 | 0.6304 ± 0.0301 |
| CNMF | 0.7841 ± 0.0451 | 0.6888 ± 0.0256 | 0.7893 ± 0.0373 | 0.6648 ± 0.0370 | 0.6561 ± 0.0415 | 0.6741 ± 0.0338 |
| GNMFLD | 0.9838 ± 0.0018 | 0.9586 ± 0.0038 | 0.9838 ± 0.0018 | 0.9680 ± 0.0035 | 0.9678 ± 0.0036 | 0.9682 ± 0.0035 |
| EDNMF | 0.8230 ± 0.0075 | 0.7131 ± 0.0069 | 0.8230 ± 0.0075 | 0.7005 ± 0.0109 | 0.6955 ± 0.0130 | 0.7056 ± 0.0093 |
| LCNMF | 0.7751 ± 0.0572 | 0.6865 ± 0.0367 | 0.7811 ± 0.0544 | 0.6641 ± 0.0513 | 0.6573 ± 0.0538 | 0.6710 ± 0.0492 |
| GOCNMF | 0.9840 ± 0.0016 | 0.9593 ± 0.0033 | 0.9840 ± 0.0016 | 0.9684 ± 0.0030 | 0.9682 ± 0.0031 | 0.9686 ± 0.0030 |
| SNMFWLP | 0.9401 ± 0.0551 | 0.9455 ± 0.0165 | 0.9459 ± 0.0474 | 0.9304 ± 0.0469 | 0.9044 ± 0.0791 | 0.9610 ± 0.0132 |
| HLPNMF | 0.9859 ± 0.0015 | 0.9632 ± 0.0031 | 0.9859 ± 0.0015 | 0.9720 ± 0.0028 | 0.9720 ± 0.0029 | 0.9721 ± 0.0028 |
| PCE_NMF | 0.9832 ± 0.0016 | 0.9579 ± 0.0033 | 0.9832 ± 0.0016 | 0.9667 ± 0.0031 | 0.9665 ± 0.0031 | 0.9669 ± 0.0030 |

### *5.3. Paired statistical analysis*

For each data set, we first averaged each method over the same 20 seeds. Across-data-set comparisons then used the ten data-set means as paired units, following the distinction between repeated trials and independent data sets in multi-algorithm comparisons [33]. We used the Friedman rank test [34] for

Table 7: Complete formal results on ORL. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

| Method | ACC | NMI | Purity | F-score | Precision | Recall |
|---|---|---|---|---|---|---|
| NMF | 0.5769 ± 0.0253 | 0.7709 ± 0.0146 | 0.6251 ± 0.0216 | 0.4296 ± 0.0269 | 0.3700 ± 0.0294 | 0.5134 ± 0.0279 |
| CNMF | 0.5747 ± 0.0337 | 0.7743 ± 0.0150 | 0.6264 ± 0.0263 | 0.4346 ± 0.0343 | 0.3730 ± 0.0404 | 0.5224 ± 0.0259 |
| GNMFLD | 0.6235 ± 0.0195 | 0.7686 ± 0.0128 | 0.6369 ± 0.0187 | 0.4438 ± 0.0325 | 0.3771 ± 0.0410 | 0.5421 ± 0.0195 |
| EDNMF | 0.5461 ± 0.0303 | 0.7010 ± 0.0214 | 0.5568 ± 0.0311 | 0.3554 ± 0.0337 | 0.3048 ± 0.0359 | 0.4274 ± 0.0309 |
| LCNMF | 0.5849 ± 0.0209 | 0.7314 ± 0.0148 | 0.5965 ± 0.0198 | 0.4063 ± 0.0285 | 0.3672 ± 0.0298 | 0.4552 ± 0.0286 |
| GOCNMF | 0.6426 ± 0.0169 | 0.7963 ± 0.0099 | 0.6644 ± 0.0184 | 0.4900 ± 0.0262 | 0.4281 ± 0.0333 | 0.5743 ± 0.0203 |
| SNMFWLP | 0.5742 ± 0.0299 | 0.7667 ± 0.0144 | 0.6124 ± 0.0250 | 0.3774 ± 0.0491 | 0.2872 ± 0.0537 | 0.5640 ± 0.0176 |
| HLPNMF | 0.5935 ± 0.0175 | 0.7904 ± 0.0079 | 0.6251 ± 0.0198 | 0.4485 ± 0.0250 | 0.3829 ± 0.0351 | 0.5437 ± 0.0124 |
| PCE_NMF | 0.6506 ± 0.0208 | 0.7910 ± 0.0117 | 0.6671 ± 0.0197 | 0.4927 ± 0.0293 | 0.4352 ± 0.0373 | 0.5693 ± 0.0215 |

Table 8: Complete formal results on PIE. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

| Method | ACC | NMI | Purity | F-score | Precision | Recall |
|---|---|---|---|---|---|---|
| NMF | 0.5380 ± 0.0167 | 0.7992 ± 0.0091 | 0.5897 ± 0.0122 | 0.4562 ± 0.0184 | 0.4129 ± 0.0222 | 0.5101 ± 0.0147 |
| CNMF | 0.5564 ± 0.0161 | 0.8020 ± 0.0082 | 0.6074 ± 0.0102 | 0.4702 ± 0.0169 | 0.4238 ± 0.0210 | 0.5283 ± 0.0121 |
| GNMFLD | 0.9416 ± 0.0061 | 0.9508 ± 0.0042 | 0.9416 ± 0.0061 | 0.8888 ± 0.0104 | 0.8826 ± 0.0119 | 0.8951 ± 0.0091 |
| EDNMF | 0.7682 ± 0.0140 | 0.8463 ± 0.0098 | 0.7717 ± 0.0142 | 0.6440 ± 0.0217 | 0.6196 ± 0.0218 | 0.6704 ± 0.0226 |
| LCNMF | 0.7565 ± 0.0095 | 0.8382 ± 0.0042 | 0.7602 ± 0.0096 | 0.6312 ± 0.0121 | 0.6126 ± 0.0149 | 0.6511 ± 0.0105 |
| GOCNMF | 0.9352 ± 0.0099 | 0.9534 ± 0.0055 | 0.9352 ± 0.0099 | 0.8829 ± 0.0147 | 0.8715 ± 0.0163 | 0.8947 ± 0.0134 |
| SNMFWLP | 0.5540 ± 0.0178 | 0.7549 ± 0.0170 | 0.6159 ± 0.0179 | 0.3067 ± 0.0286 | 0.2114 ± 0.0248 | 0.5622 ± 0.0255 |
| HLPNMF | 0.9589 ± 0.0088 | 0.9714 ± 0.0049 | 0.9589 ± 0.0088 | 0.9267 ± 0.0142 | 0.9186 ± 0.0168 | 0.9350 ± 0.0117 |
| PCE_NMF | 0.9376 ± 0.0110 | 0.9527 ± 0.0068 | 0.9376 ± 0.0110 | 0.8856 ± 0.0179 | 0.8761 ± 0.0201 | 0.8953 ± 0.0158 |

Table 9: Complete formal results on COIL100. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

| Method | ACC | NMI | Purity | F-score | Precision | Recall |
|---|---|---|---|---|---|---|
| NMF | 0.4620 ± 0.0123 | 0.7389 ± 0.0040 | 0.5038 ± 0.0111 | 0.4114 ± 0.0084 | 0.3594 ± 0.0105 | 0.4811 ± 0.0083 |
| CNMF | 0.4776 ± 0.0178 | 0.7450 ± 0.0050 | 0.5175 ± 0.0142 | 0.4221 ± 0.0120 | 0.3714 ± 0.0146 | 0.4891 ± 0.0112 |
| GNMFLD | 0.8177 ± 0.0065 | 0.8945 ± 0.0035 | 0.8181 ± 0.0065 | 0.7413 ± 0.0087 | 0.7259 ± 0.0109 | 0.7574 ± 0.0081 |
| EDNMF | 0.5549 ± 0.0104 | 0.7233 ± 0.0061 | 0.5584 ± 0.0100 | 0.4366 ± 0.0100 | 0.3996 ± 0.0121 | 0.4814 ± 0.0092 |
| LCNMF | 0.5442 ± 0.0118 | 0.7188 ± 0.0061 | 0.5480 ± 0.0111 | 0.4288 ± 0.0102 | 0.3887 ± 0.0111 | 0.4782 ± 0.0109 |
| GOCNMF | 0.8166 ± 0.0076 | 0.8731 ± 0.0062 | 0.8166 ± 0.0076 | 0.6997 ± 0.0127 | 0.6865 ± 0.0123 | 0.7133 ± 0.0135 |
| SNMFWLP | 0.4131 ± 0.0178 | 0.6272 ± 0.0167 | 0.4670 ± 0.0158 | 0.1756 ± 0.0221 | 0.1161 ± 0.0180 | 0.3643 ± 0.0195 |
| HLPNMF | 0.8740 ± 0.0067 | 0.9229 ± 0.0039 | 0.8743 ± 0.0064 | 0.7961 ± 0.0100 | 0.7803 ± 0.0125 | 0.8126 ± 0.0086 |
| PCE_NMF | 0.9100 ± 0.0058 | 0.9452 ± 0.0030 | 0.9101 ± 0.0058 | 0.8590 ± 0.0090 | 0.8505 ± 0.0110 | 0.8677 ± 0.0072 |

Table 10: Complete formal results on AR. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

| Method | ACC | NMI | Purity | F-score | Precision | Recall |
|---|---|---|---|---|---|---|
| NMF | 0.6094 ± 0.0192 | 0.8413 ± 0.0111 | 0.6594 ± 0.0166 | 0.4966 ± 0.0259 | 0.4326 ± 0.0271 | 0.5833 ± 0.0248 |
| CNMF | 0.5919 ± 0.0149 | 0.8327 ± 0.0082 | 0.6426 ± 0.0134 | 0.4786 ± 0.0198 | 0.4159 ± 0.0208 | 0.5640 ± 0.0216 |
| GNMFLD | 0.6142 ± 0.0137 | 0.8242 ± 0.0049 | 0.6279 ± 0.0126 | 0.4763 ± 0.0152 | 0.4240 ± 0.0204 | 0.5437 ± 0.0084 |
| EDNMF | 0.6355 ± 0.0220 | 0.8076 ± 0.0146 | 0.6393 ± 0.0222 | 0.4695 ± 0.0293 | 0.4032 ± 0.0325 | 0.5628 ± 0.0238 |
| LCNMF | 0.6902 ± 0.0131 | 0.8409 ± 0.0070 | 0.6971 ± 0.0135 | 0.5505 ± 0.0164 | 0.5106 ± 0.0184 | 0.5974 ± 0.0174 |
| GOCNMF | 0.6323 ± 0.0097 | 0.8394 ± 0.0038 | 0.6485 ± 0.0086 | 0.4889 ± 0.0134 | 0.4249 ± 0.0186 | 0.5762 ± 0.0083 |
| SNMFWLP | 0.5097 ± 0.0136 | 0.7761 ± 0.0092 | 0.5353 ± 0.0135 | 0.2742 ± 0.0320 | 0.1896 ± 0.0284 | 0.5040 ± 0.0183 |
| HLPNMF | 0.6390 ± 0.0148 | 0.8418 ± 0.0056 | 0.6515 ± 0.0136 | 0.5073 ± 0.0172 | 0.4494 ± 0.0218 | 0.5828 ± 0.0136 |
| PCE_NMF | 0.6332 ± 0.0098 | 0.8381 ± 0.0041 | 0.6481 ± 0.0091 | 0.4892 ± 0.0148 | 0.4245 ± 0.0200 | 0.5776 ± 0.0084 |

Table 11: Complete formal results on MSRA25. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

| Method | ACC | NMI | Purity | F-score | Precision | Recall |
|---|---|---|---|---|---|---|
| NMF | 0.4906 ± 0.0183 | 0.5455 ± 0.0161 | 0.5084 ± 0.0153 | 0.3777 ± 0.0137 | 0.3542 ± 0.0146 | 0.4048 ± 0.0160 |
| CNMF | 0.5125 ± 0.0291 | 0.5648 ± 0.0246 | 0.5293 ± 0.0229 | 0.4000 ± 0.0237 | 0.3785 ± 0.0242 | 0.4241 ± 0.0243 |
| GNMFLD | 0.9904 ± 0.0029 | 0.9814 ± 0.0048 | 0.9904 ± 0.0029 | 0.9813 ± 0.0058 | 0.9807 ± 0.0052 | 0.9818 ± 0.0064 |
| EDNMF | 0.5844 ± 0.0530 | 0.5337 ± 0.0361 | 0.5942 ± 0.0483 | 0.4352 ± 0.0461 | 0.4218 ± 0.0471 | 0.4501 ± 0.0479 |
| LCNMF | 0.5434 ± 0.0327 | 0.4991 ± 0.0234 | 0.5529 ± 0.0292 | 0.3752 ± 0.0265 | 0.3493 ± 0.0319 | 0.4061 ± 0.0224 |
| GOCNMF | 0.9748 ± 0.0100 | 0.9588 ± 0.0133 | 0.9748 ± 0.0100 | 0.9514 ± 0.0192 | 0.9494 ± 0.0198 | 0.9533 ± 0.0188 |
| SNMFWLP | 0.9269 ± 0.0451 | 0.9630 ± 0.0151 | 0.9450 ± 0.0323 | 0.9283 ± 0.0397 | 0.8963 ± 0.0617 | 0.9641 ± 0.0189 |
| HLPNMF | 0.9941 ± 0.0014 | 0.9878 ± 0.0027 | 0.9941 ± 0.0014 | 0.9881 ± 0.0028 | 0.9875 ± 0.0030 | 0.9887 ± 0.0028 |
| PCE_NMF | 0.9788 ± 0.0082 | 0.9607 ± 0.0126 | 0.9788 ± 0.0082 | 0.9593 ± 0.0154 | 0.9595 ± 0.0142 | 0.9592 ± 0.0166 |

Table 12: Complete formal results on MNIST. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

| Method | ACC | NMI | Purity | F-score | Precision | Recall |
|---|---|---|---|---|---|---|
| NMF | 0.4851 ± 0.0262 | 0.4524 ± 0.0144 | 0.5341 ± 0.0228 | 0.3876 ± 0.0169 | 0.3720 ± 0.0186 | 0.4047 ± 0.0163 |
| CNMF | 0.5069 ± 0.0281 | 0.4683 ± 0.0182 | 0.5588 ± 0.0252 | 0.4083 ± 0.0217 | 0.3916 ± 0.0233 | 0.4269 ± 0.0223 |
| GNMFLD | 0.9324 ± 0.0033 | 0.8528 ± 0.0047 | 0.9324 ± 0.0033 | 0.8729 ± 0.0057 | 0.8700 ± 0.0062 | 0.8758 ± 0.0051 |
| EDNMF | 0.5557 ± 0.0181 | 0.4577 ± 0.0092 | 0.5767 ± 0.0173 | 0.4132 ± 0.0112 | 0.4098 ± 0.0114 | 0.4167 ± 0.0111 |
| LCNMF | 0.5400 ± 0.0223 | 0.4490 ± 0.0103 | 0.5636 ± 0.0175 | 0.4064 ± 0.0148 | 0.4017 ± 0.0153 | 0.4113 ± 0.0147 |
| GOCNMF | 0.9352 ± 0.0027 | 0.8571 ± 0.0041 | 0.9352 ± 0.0027 | 0.8778 ± 0.0047 | 0.8755 ± 0.0051 | 0.8802 ± 0.0043 |
| SNMFWLP | 0.9342 ± 0.0026 | 0.8576 ± 0.0038 | 0.9342 ± 0.0026 | 0.8759 ± 0.0046 | 0.8734 ± 0.0051 | 0.8784 ± 0.0042 |
| HLPNMF | 0.9291 ± 0.0040 | 0.8508 ± 0.0046 | 0.9291 ± 0.0040 | 0.8677 ± 0.0066 | 0.8652 ± 0.0074 | 0.8702 ± 0.0058 |
| PCE_NMF | 0.9315 ± 0.0034 | 0.8501 ± 0.0052 | 0.9315 ± 0.0034 | 0.8711 ± 0.0058 | 0.8685 ± 0.0062 | 0.8736 ± 0.0054 |

Table 13: Complete formal results on USPS. Entries are mean ± sample standard deviation over 20 paired seeds, evaluated on unlabeled samples.

| Method | ACC | NMI | Purity | F-score | Precision | Recall |
|---|---|---|---|---|---|---|
| NMF | 0.6671 ± 0.0305 | 0.6092 ± 0.0196 | 0.7142 ± 0.0264 | 0.5790 ± 0.0227 | 0.5874 ± 0.0245 | 0.5710 ± 0.0214 |
| CNMF | 0.6736 ± 0.0423 | 0.6108 ± 0.0188 | 0.7139 ± 0.0240 | 0.5806 ± 0.0253 | 0.5884 ± 0.0260 | 0.5731 ± 0.0252 |
| GNMFLD | 0.9685 ± 0.0010 | 0.9202 ± 0.0020 | 0.9685 ± 0.0010 | 0.9440 ± 0.0017 | 0.9389 ± 0.0019 | 0.9491 ± 0.0016 |
| EDNMF | 0.7713 ± 0.0022 | 0.6606 ± 0.0041 | 0.7713 ± 0.0022 | 0.6515 ± 0.0035 | 0.6523 ± 0.0045 | 0.6507 ± 0.0030 |
| LCNMF | 0.7652 ± 0.0192 | 0.6584 ± 0.0117 | 0.7677 ± 0.0093 | 0.6480 ± 0.0136 | 0.6527 ± 0.0126 | 0.6434 ± 0.0148 |
| GOCNMF | 0.9695 ± 0.0012 | 0.9222 ± 0.0024 | 0.9695 ± 0.0012 | 0.9459 ± 0.0021 | 0.9414 ± 0.0022 | 0.9504 ± 0.0020 |
| SNMFWLP | 0.9700 ± 0.0012 | 0.9235 ± 0.0023 | 0.9700 ± 0.0012 | 0.9473 ± 0.0020 | 0.9438 ± 0.0021 | 0.9507 ± 0.0020 |
| HLPNMF | 0.9719 ± 0.0011 | 0.9268 ± 0.0026 | 0.9719 ± 0.0011 | 0.9503 ± 0.0020 | 0.9474 ± 0.0022 | 0.9531 ± 0.0019 |
| PCE_NMF | 0.9700 ± 0.0011 | 0.9226 ± 0.0024 | 0.9700 ± 0.0011 | 0.9465 ± 0.0021 | 0.9424 ± 0.0024 | 0.9506 ± 0.0020 |

the multi-method comparison and the paired Wilcoxon test [35] with Holm adjustment [36] for the selected pairwise comparisons. The mean PCE-NMF minus HLPNMF ACC difference was 0.0031 (four wins in ten data sets; Holm-adjusted Wilcoxon $p = 0.9219$). Against GOCNMF, the difference was 0.0150 (seven wins; adjusted $p = 0.4648$). The corresponding NMI differences were $-0.0050$ and 0.0121, with adjusted $p = 0.5508$ and $p = 0.8457$. The three methods have data-dependent performance profiles. Figure 3 summarizes the nine methods' average ranks across the six metrics.

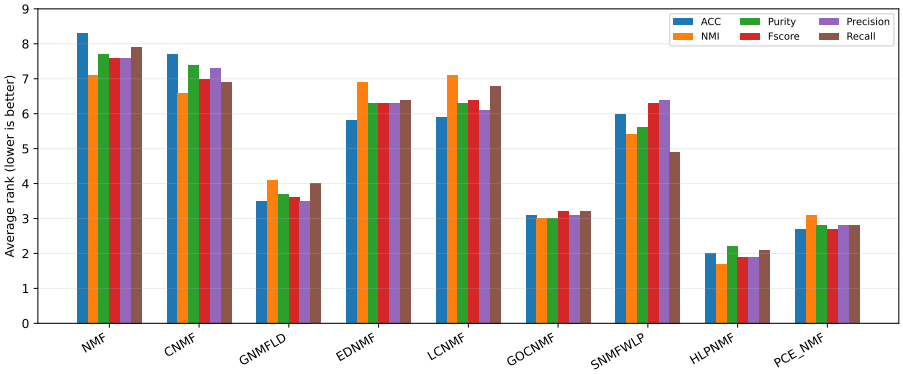



Figure 3: Average ranks of nine methods across ten data sets for six unlabeled-sample clustering metrics. Within each data set, ranking uses the method mean over 20 seeds; lower average ranks are better.

### *5.4. Controlled mechanism comparison*

We compared PCE-NMF with HCE-NMF, which retains NMF and the labeled support constraint but removes propagated exclusion, and with a harmonic-propagation-only predictor using the same graph field. PCE-NMF exceeded HCE-NMF in ACC on all ten data sets. The mean data-set-level difference was 0.4319 (95% confidence interval 0.3124–0.5514; Holm-adjusted $p = 0.0039$). Against harmonic propagation alone, its mean difference was $-0.0006$ (95% confidence interval $-0.0134$–$0.0121$; adjusted $p = 0.6953$). Propagated exclusion therefore contributed substantially relative to hard exclusion in this controlled comparison.

Table 14 resolves the aggregate comparison by data set. The factor-based rule exceeds harmonic prediction on COIL100 by 0.0243 and on COIL20 by 0.0143, whereas harmonic prediction exceeds it on YaleB by 0.0428 and on PIE by 0.0113. Thus the two decisions respond differently to the same graph evidence, even though their mean ACC across the ten data sets is close.

### *5.5. Parameter sensitivity and labeled-sample proportion*

PCE-NMF parameters were selected on five development splits separate from the formal evaluation. The exclusion weight was assessed before the graph-neighborhood size; data sets without an inner validation sample per class used the predeclared transfer rule. Figure 4 shows the recorded validation ACC for four representative data sets. The orange points mark locked choices; the graph-size sweep uses each data set's selected exclusion weight.

Table 14: Mean unlabeled-sample ACC over 20 paired seeds in the mechanism controls. HCE-NMF uses hard labeled support without propagated exclusion; harmonic-only predicts directly from the same graph field used by PCE-NMF.

| Data set | PCE-NMF | HCE-NMF | Harmonic-only |
|---|---|---|---|
| YaleB | 0.6934 | 0.1469 | 0.7362 |
| COIL20 | 0.9713 | 0.6606 | 0.9570 |
| Optdigits | 0.9832 | 0.7041 | 0.9825 |
| ORL | 0.6506 | 0.3011 | 0.6418 |
| PIE | 0.9376 | 0.3292 | 0.9489 |
| COIL100 | 0.9100 | 0.3795 | 0.8857 |
| AR | 0.6332 | 0.2726 | 0.6297 |
| MSRA25 | 0.9788 | 0.2298 | 0.9783 |
| MNIST | 0.9315 | 0.5771 | 0.9353 |
| USPS | 0.9700 | 0.7396 | 0.9702 |

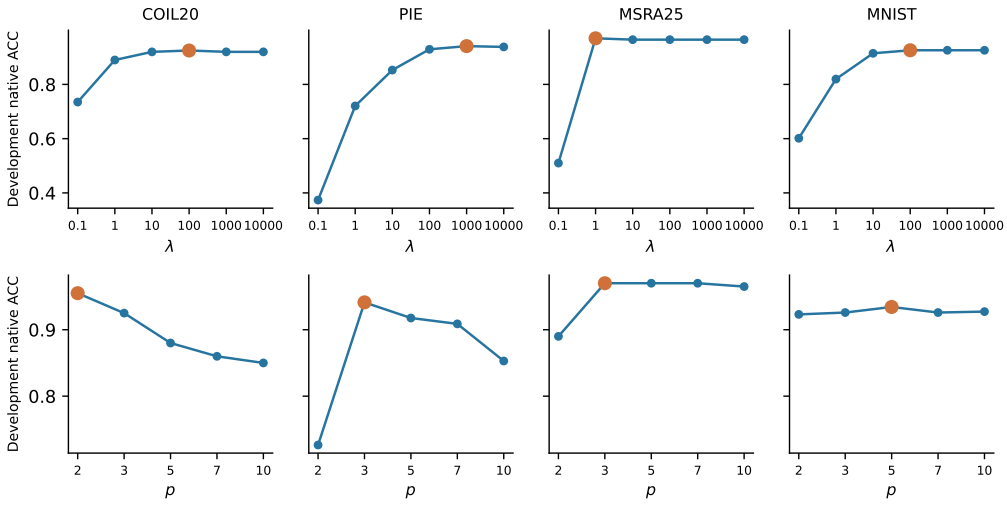



Figure 4: PCE-NMF parameter development on COIL20, PIE, MSRA25, and MNIST. Mean native validation ACC over five development splits. Orange points mark locked parameters. The upper row varies exclusion weight $\lambda$; the lower row varies graph neighborhood size $p$ at the selected $\lambda$.

We then evaluated 5%, 10%, 15%, 20%, and 30% labeled samples per class on PIE, MSRA25, COIL20, and MNIST. The 10% points reuse the main-comparison trials. At 5% labels, PCE-NMF obtained ACC 0.9459 on COIL20 versus 0.8101 for HLPNMF. On PIE and MSRA25, HLPNMF remained higher at the same ratio (0.9094 versus 0.8854, and 0.9770 versus 0.9460). Figure 5 shows the complete preselected range.

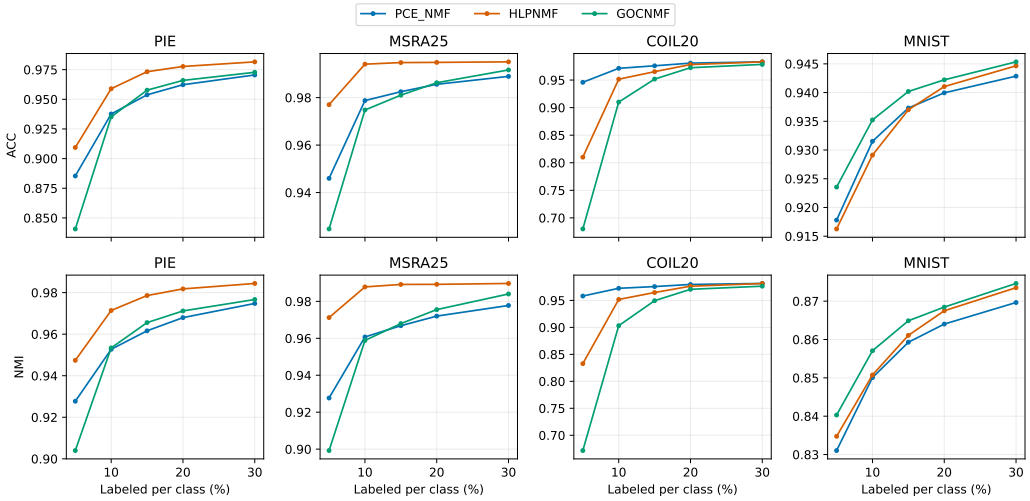



Figure 5: Mean ACC and NMI over 20 seeds at five labeled-sample proportions on four preselected data sets. The 10% values are the audited main-comparison results.

### *5.6. Noise robustness*

*Incorrect observed labels..* We also changed 0–3 observed labels per class on COIL20 and PIE while keeping the labeled indices, data, initialization, and evaluation labels fixed within each seed. All nine methods received the same altered labels. At three incorrect labels per class, the mean paired ACC changes from the clean condition were −0.3765 for PCE-NMF, −0.3602 for HLPNMF, and −0.3812 for GOCNMF on COIL20; on PIE they were −0.5558, −0.5524, and −0.5647, respectively. The three methods show similar degradation at this corruption level. Figure 6 shows all preselected levels.

We corrupted 10%–50% of samples using either salt-and-pepper noise or a random $5 \times 5$ black occlusion, with the clean and corrupted trials sharing labeled splits. Figure 7 illustrates the two pixel-level perturbations on representative samples from the three image data sets.

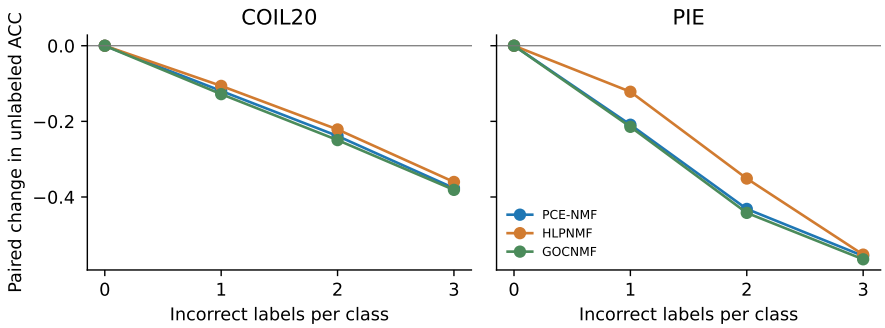



Figure 6: Mean paired change in unlabeled-sample ACC under 0–3 incorrect observed labels per class on COIL20 and PIE. Each point uses 20 matched seeds; the same altered labels are supplied to all methods within each condition.

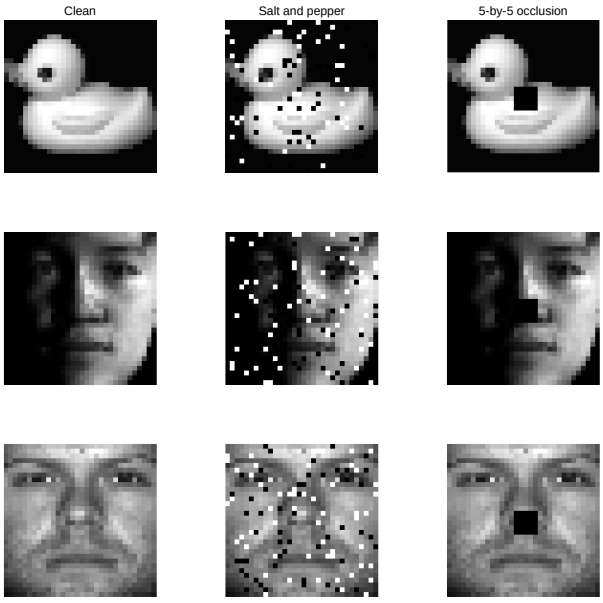



Figure 7: Representative 32-by-32 samples from COIL20, PIE, and YaleB before and after salt-and-pepper corruption (pixel probability 0.1) or a 5-by-5 black occlusion. Each row uses a common grayscale range. The images show the applied input perturbations.

Across the three selected data sets and five corruption levels, the descriptive mean ACC changes under salt-and-pepper noise were −0.0497 for PCE-NMF, −0.0685 for HLPNMF, and −0.0414 for GOCNMF. Under occlusion, they

were −0.0146, −0.0107, and −0.0061, respectively. The relative degradation depends on the corruption type and data set; Figure 8 displays the paired changes by condition.

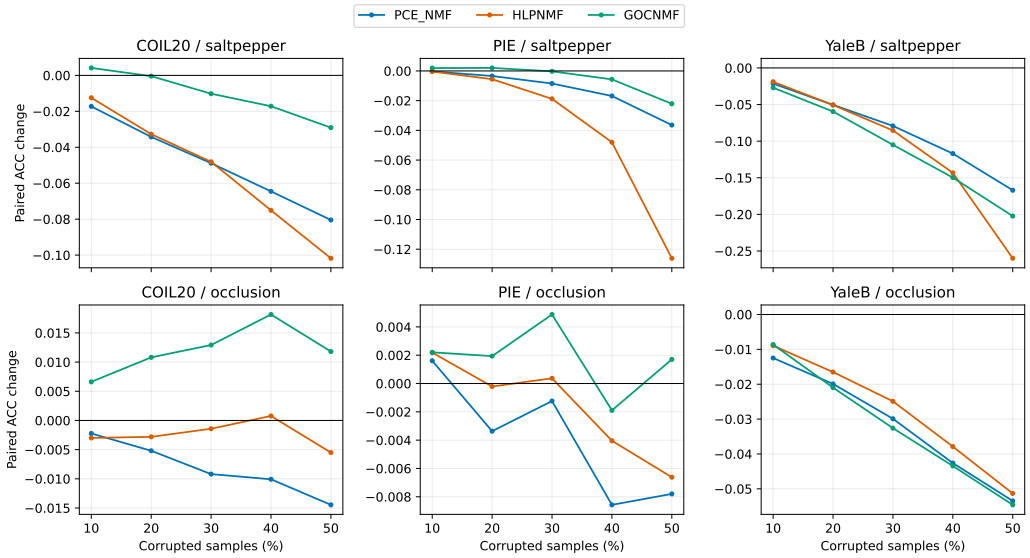



Figure 8: Mean paired change in unlabeled-sample ACC from the clean trial under salt-and-pepper noise and $5 \times 5$ occlusion. Each point uses 20 matched seeds. Columns identify the data set and rows the corruption type.

### *5.7. Scalability and running time*

On balanced COIL100 subsets containing 20, 40, 60, 80, and 100 classes, PCE-NMF had the highest mean ACC among PCE-NMF, HLPNMF, and GOCNMF at every tested size. At 100 classes, their ACC means were 0.9100, 0.8740, and 0.8166. The corresponding measured wall times were 14.274, 123.278, and 13.254 seconds, respectively. Figure 9 shows performance and measured time as the selected COIL100 subset grows.

In a separate ten-data-set timing study, each trial rebuilt its graph and timed construction, fitting, and native prediction. PCE-NMF was faster than HLPNMF and slower than GOCNMF on every data set. On COIL100, their respective mean times were 10.752, 92.622, and 9.485 seconds. Figure 10 reports all nine methods on the same machine and implementation.

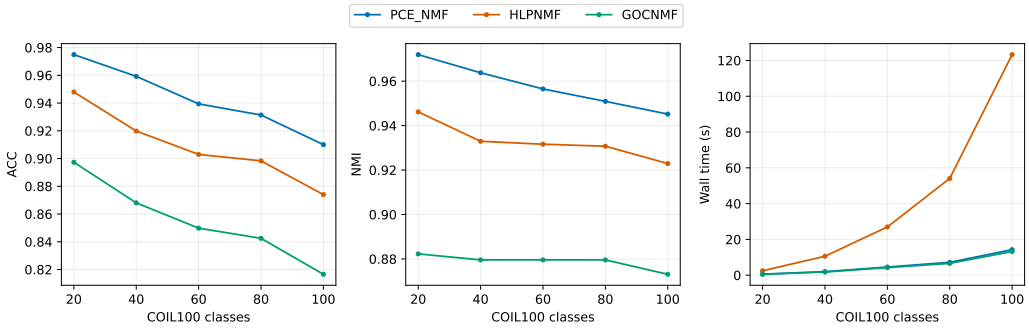



Figure 9: ACC, NMI, and measured wall time on balanced COIL100 subsets containing 20–100 classes. Each point averages 20 seeds.

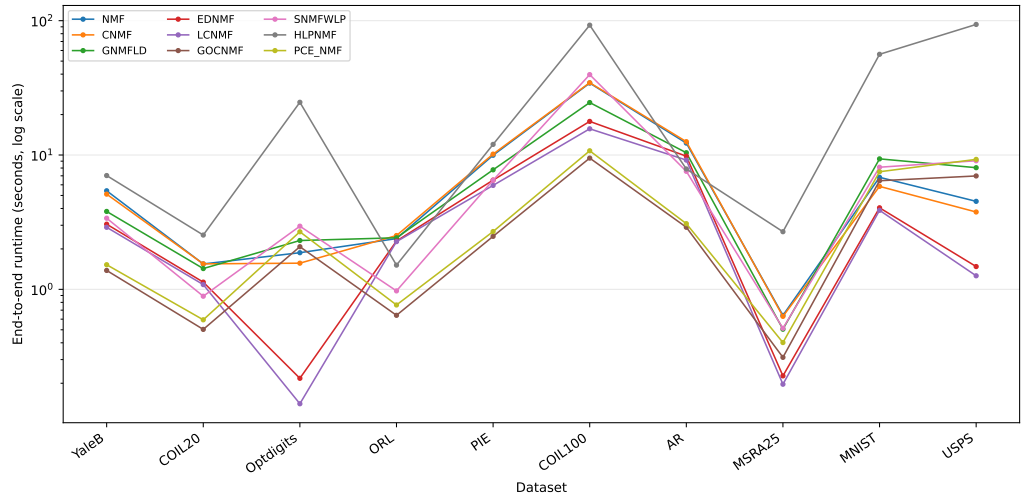



Figure 10: Mean end-to-end runtime over 20 seeds for nine methods on ten data sets, including each run's graph construction, fitting, and native prediction. No persistent graph cache was shared across timed trials. The vertical axis is logarithmic.

*5.8. Numerical convergence*

We recorded the original PCE-NMF objective at initialization and after each of its 50 updates in one formal run on YaleB, COIL20, PIE, and COIL100. The objective decreased at every recorded step on all four data sets. All values remained finite, and the measured nonnegativity and labeled-support violations were zero. Figure 11 shows these objective trajectories.

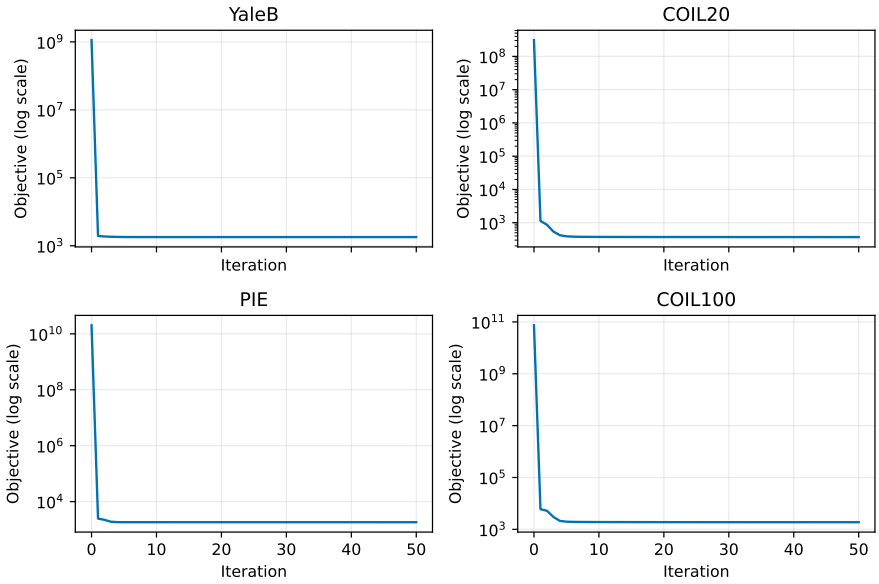



Figure 11: PCE-NMF objective trajectories over 50 updates on four representative data sets using one formal run and locked parameters. The objective axis is logarithmic.

## 6. Discussion

The controls measure two effects separately. The large difference from HCE-NMF shows the contribution of propagated exclusion within the factorization. Direct harmonic prediction has nearly the same mean ACC as PCE-NMF across the ten data sets. The factor-based rule gains on COIL20 and COIL100 and loses on YaleB, locating its empirical effect at the data-set level. The harmonic field supplies a strong direct predictor as well as the exclusion weights used in PCE-NMF.

PCE-NMF retains a non-negative class-aware representation and an explicit reconstruction term alongside direct class decisions. HLPNMF achieves

higher ACC on several data sets, whereas PCE-NMF is faster in the measured no-cache pipeline. The graph construction is quadratic in sample count before neighbor selection, so the measured runtime comparison includes this graph cost. The class-count experiment characterizes the tested COIL100 subsets; its accuracy ordering is specific to those subsets.

## 7. Conclusion

PCE-NMF turns harmonic class evidence into penalties on class-wise NMF contributions and uses those contributions for direct decisions. Its class-energy rule is invariant to factor rescaling. Across ten data sets, propagated exclusion improved on a matched hard-support model. Against direct harmonic prediction, its mean ACC difference was $-0.0006$, with gains and losses across individual data sets. PCE-NMF performed strongly on COIL20 and COIL100 and ran faster than HLPNMF, while HLPNMF achieved higher ACC on several other data sets. The combined results define the method's mathematical construction and its measured performance profile under matched semi-supervised evaluation.

## CRediT authorship contribution statement

Deshu Sun: Writing-original draft, Methodology, Software, Conceptualization, Formal analysis. Feng Wang: Writing-review and editing, Funding acquisition, Visualization, Conceptualization. All authors reviewed the manuscript.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data and code availability

Data will be made available on request.

**Acknowledgments**

This work was supported by Guizhou Provincial Science and Technology Projects (2025039, 2026283), the High-Level Innovative Talent Project of Guizhou Province (GCC2023027), the Natural Science Research Project of Department of Education of Guizhou Province (2024259), and the Research Foundation of Guizhou Minzu University (GZMUZK[2023]YB10).

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